Zorian Thornton

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Professional Summary

Computational biologist and machine learning researcher with 6+ years of experience developing predictive models for biological systems. Expertise in building end-to-end deep learning pipelines for large-scale sequence analysis, implementing Bayesian statistical methods, and translating complex biological data into actionable insights. Proven track record of developing production-grade tools for viral evolution modeling and protein phenotype prediction.

Selected Projects

Protein Language Model Portfolio | github.com/zorian15/bio-transformer-portfolio 2026

- Benchmarked ESM-2 zero-shot scoring, linear probes, and LoRA fine-tuning against structured baselines on ProteinGym deep mutational scanning assays, reproducing published ProteinGym results to validate the pipeline
- Established that zero-shot masked-marginal scoring is additive by construction and therefore cannot represent epistasis, motivating supervised comparison against interpretable biophysical models on multi-mutant SARS-CoV-2 RBD data
- Measured the effect of concatenating functional-annotation text embeddings onto ESM-2 representations for subcellular localization prediction, raising macro-F1 from 0.616 to 0.740 across 2,773 homology-partitioned held-out proteins; shuffled-text and structured-field controls bound how much of that gain could be label leakage rather than functional signal
- Built a shared, typed, tested Python package (pytest, mypy, ruff, GitHub Actions CI including an embedding-drift anchor), a published documentation site, and SLURM/CUDA job templates

Technical Skills

Languages: Python, R, SQL, Java, MATLAB, bash
Machine Learning: PyTorch, TensorFlow, JAX, NumPyro, scikit-learn, supervised/unsupervised learning, deep learning, transfer learning, Bayesian inference
Protein ML: Protein language models (ESM-2), LoRA / parameter-efficient fine-tuning, Hugging Face, embedding-based property prediction, deep mutational scanning analysis
Protein Design & Structure: Rosetta, RoseTTAFold (graduate coursework with D. Baker and F. DiMaio); protein–protein interaction and interface analysis
Statistical Methods: MCMC, variational inference, hierarchical modeling, dimensionality reduction, model benchmarking
Bioinformatics & Computational Biology: Phylogenetic inference, sequence analysis, antigenic evolution modeling, Augur, Auspice, Biopython, evofr
Data Science & Visualization: pandas, numpy, scipy, matplotlib, seaborn, statistical modeling
Software Engineering: Git, pytest, mypy, ruff, GitHub Actions CI, code review, SLURM, CUDA, parallel computing, AWS, Redshift, Unix/Linux, high-performance computing

Research & Work Experience

Genentech South San Francisco, CA
Data Science Intern, Evidence for Access (Technical Institute) *June 2026 – September 2026*

- Implemented and deployed a machine learning model (CatBoost) for imputation of proxied costs on 100M+ claims records, supporting Health Economics Outcomes Research (HEOR) and Real-World Data (RWD) analyses

Fred Hutchinson Cancer Research Center Seattle, WA
Doctoral Researcher, Computational Biology (Advisor: Frederick Matsen IV) *July 2021 – June 2026*

- Built `torchdms`, a PyTorch package of interpretable neural networks predicting mutational effects on binding, expression, and stability from deep mutational scanning data across tens of thousands of protein variants
- Co-developed `polyclonal`, a biophysical model of viral escape from polyclonal antibody serum that decomposes escape into epitope-specific contributions and per-mutation effects; used in subsequent Bloom lab escape experiments
- Created `antigen-prime`, an open-source Java simulator of joint sequence and antigenic evolution of viral populations under host immune selection, used to benchmark variant-assignment and fitness-estimation methods
- Collaborated with experimental biology teams to validate computational predictions, resulting in 2 peer-reviewed publications and tools adopted by multiple research groups

Adaptive Biotechnologies Seattle, WA
Machine Learning Computational Biology Intern *June 2022 – Sept. 2022*

- Implemented and benchmarked semi-supervised learning methods for T-cell receptor specificity classification on high-throughput sequencing datasets of millions of sequences, using systematic downsampling experiments to identify optimal training set sizes across classification tasks
- Designed scalable Python preprocessing pipelines optimized for memory and runtime, and visualization dashboards to communicate findings to cross-functional teams of immunologists, clinicians, and product managers

Virginia Tech Department of Statistics Blacksburg, VA
Undergraduate Researcher (Advisor: Leah Johnson) *Aug. 2017 – May 2019*

- Developed predictive environmental models integrating temperature-dependent biological parameters with disease transmission dynamics to forecast Bluetongue virus spread across climate scenarios
- Built Bayesian hierarchical models using MCMC methods to quantify uncertainty in environmental-disease relationships, incorporating vector life history traits and host-pathogen interactions

Nielsen Chicago, IL
Data Science Intern *June 2018 – Aug. 2018*

- Implemented statistical framework to identify and correct errors in scanned receipt data, building automated pipeline for data quality assurance
- Developed anomaly detection algorithms to flag inconsistencies in consumer purchase data at scale

University of Washington Department of Genome Sciences Seattle, WA
Undergraduate Researcher (Advisor: Jim Bruce) *June 2017 – Aug. 2017*

- Built probabilistic models to quantify protein-protein interaction competition in cross-linking mass spectrometry data
- Developed 3D visualization tools for analyzing inter-surface regions of cross-linked proteins, contributing to published proteomics software (Journal of Proteome Research)

Education

University of Washington Seattle, WA
Ph.D. in Genome Sciences *Sep. 2019 – March 2026*

- Dissertation: *Computational tools for studying viral antigenic evolution*

Virginia Tech Blacksburg, VA
B.Sc. Statistics & B.Sc. Computational Modeling and Data Analytics *Aug. 2015 – May 2019*

- Minors in Mathematics and Computer Science

Selected Publications

Thornton, Z., Tran, T., Figgins, M., et al. (2026). Simulating the genetic and antigenic evolution of viruses under selection pressure from host immunity. *Manuscript under revision in Virus Evolution.*

Yu, T.*, **Thornton, Z.***, Hannon, W., et al. (2022). A biophysical model of viral escape from polyclonal antibodies. *Virus Evolution*, 8(2), veac110.

El Moustaid, F., **Thornton, Z.**, et al. (2021). Predicting temperature-dependent transmission suitability of bluetongue virus in livestock. *Parasites & Vectors*, 14(1), pp.1-14.

Keller, A., Chavez, J.D., Eng, J.K., **Thornton, Z.** and Bruce, J.E. (2018). Tools for 3D Interactome Visualization. *Journal of Proteome Research*, 18(2), pp.753-758.

Selected Presentations

Thornton, Z. Predicting Disease Progression of Colorectal Cancer via Self-Organizing Maps. RECOMB 2019, George Washington University, Washington DC, May 2019.

Thornton, Z. Modeling Bluetongue Virus via Markov Chain Monte Carlo Methods. Student Experiential Learning Conference, Virginia Tech, Blacksburg, VA, Apr. 2018.

Thornton, Z. Viewing Molecular Interaction Interfaces Through Directed Computational Methods. Department of Genome Sciences Research Symposium, University of Washington, Seattle, WA, Aug. 2017.

Honors & Awards

National Science Foundation Graduate Research Fellowship Program, Honorable Mention	2020
Mu Sigma Rho, National Honor Society for Statistics	2019
College of Science Dean's Roundtable Scholarship, Virginia Tech	2018–2019
Luther and Alice Hamlett Undergraduate Research Grant, Virginia Tech	2018–2019
Fralin Undergraduate Research Fellowship, Virginia Tech	2017–2018
Eckert Statistics Scholar, Virginia Tech	2016–2019